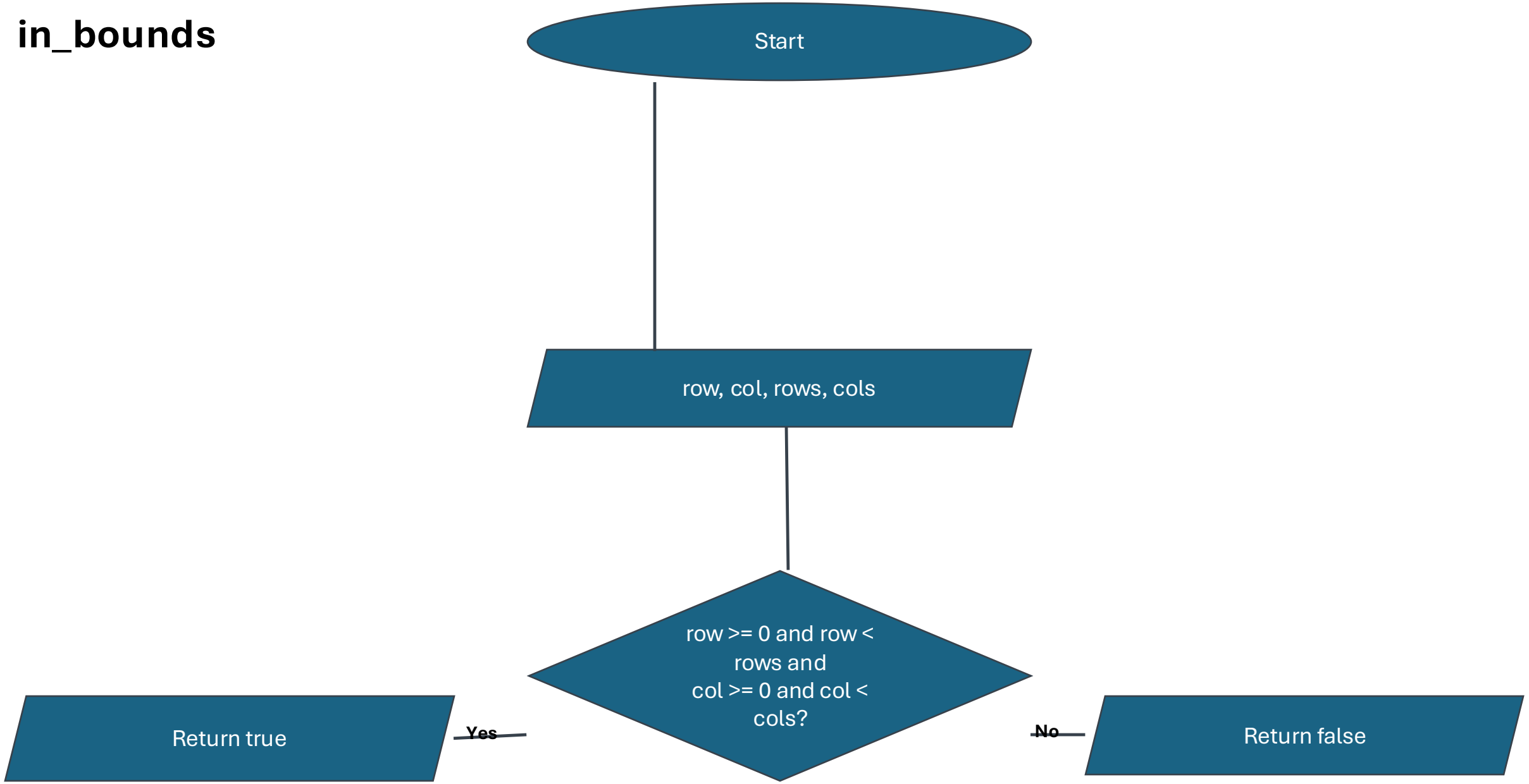
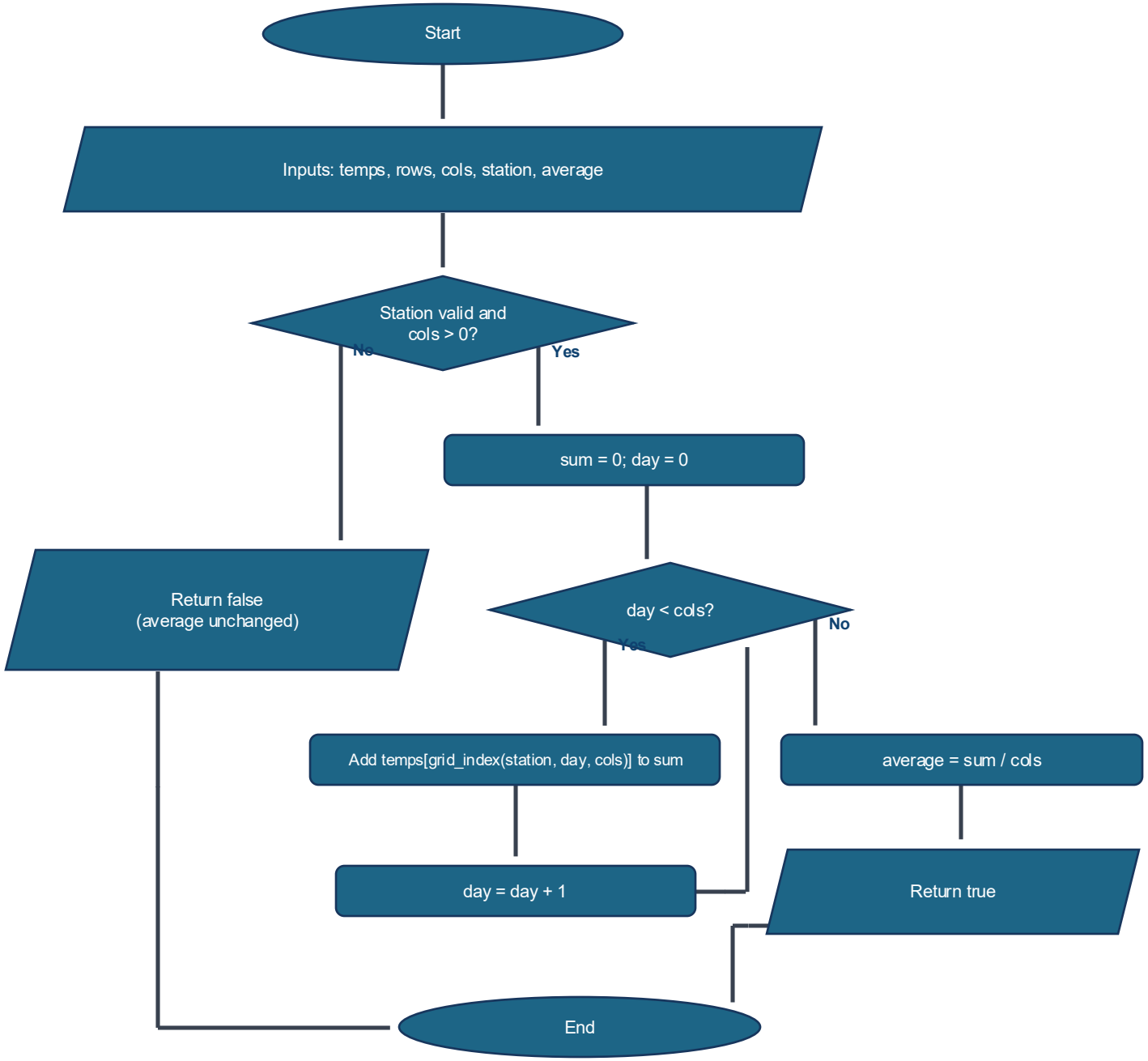
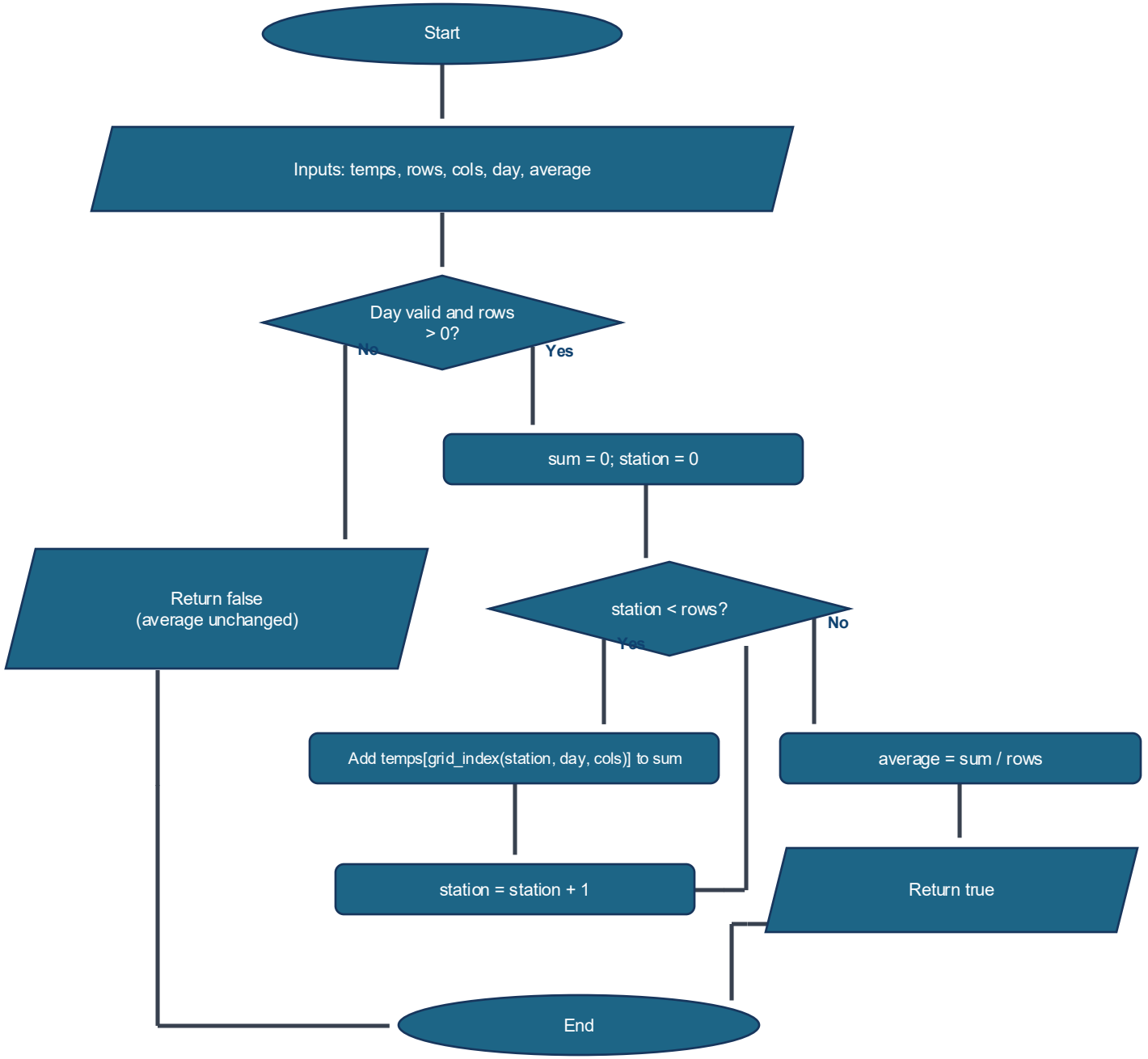
in_bounds


Start
row, col, rows, cols
row >= 0 and row < rows and col >= 0 and col < cols?
Yes
No
Return true
Return false

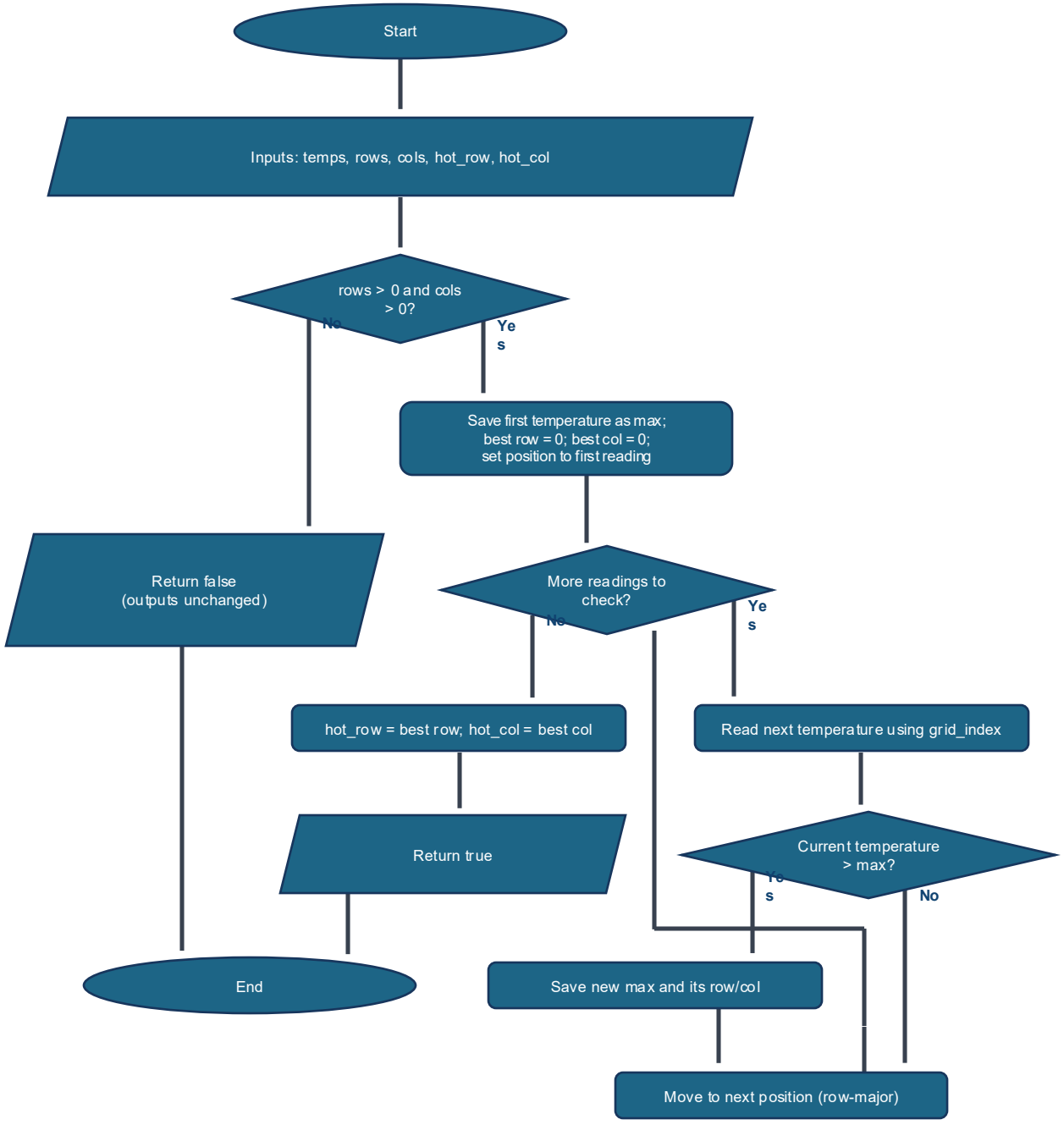
station_average



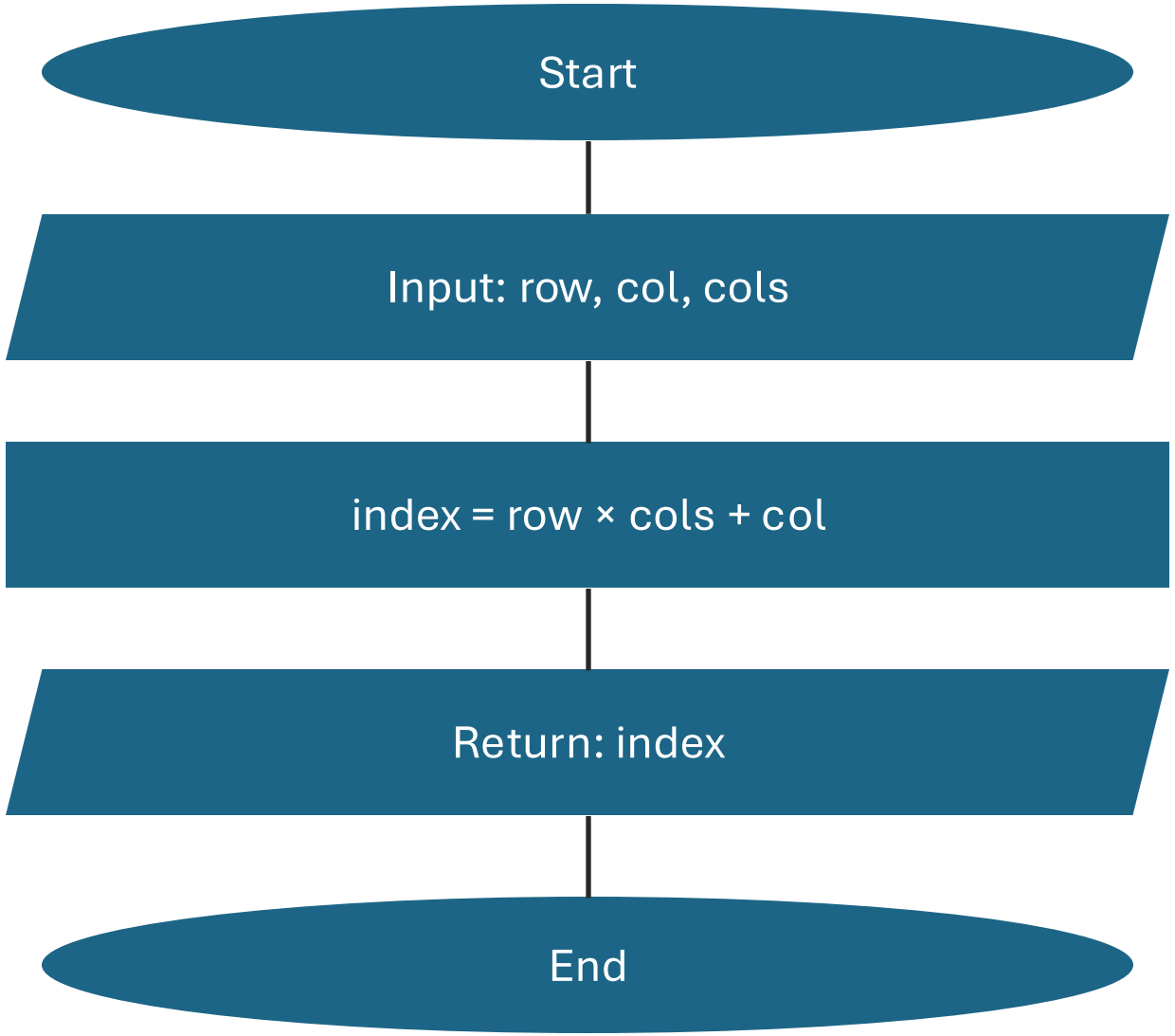
day_average


Start
Inputs: temps, rows, cols, day, average
Day valid and rows > 0?
No
Yes
sum = 0; station = 0
Return false (average unchanged)
station < rows?
Yes
No
Add temps[grid_index(station, day, cols)] to sum
average = sum / rows
station = station + 1
Return true
End

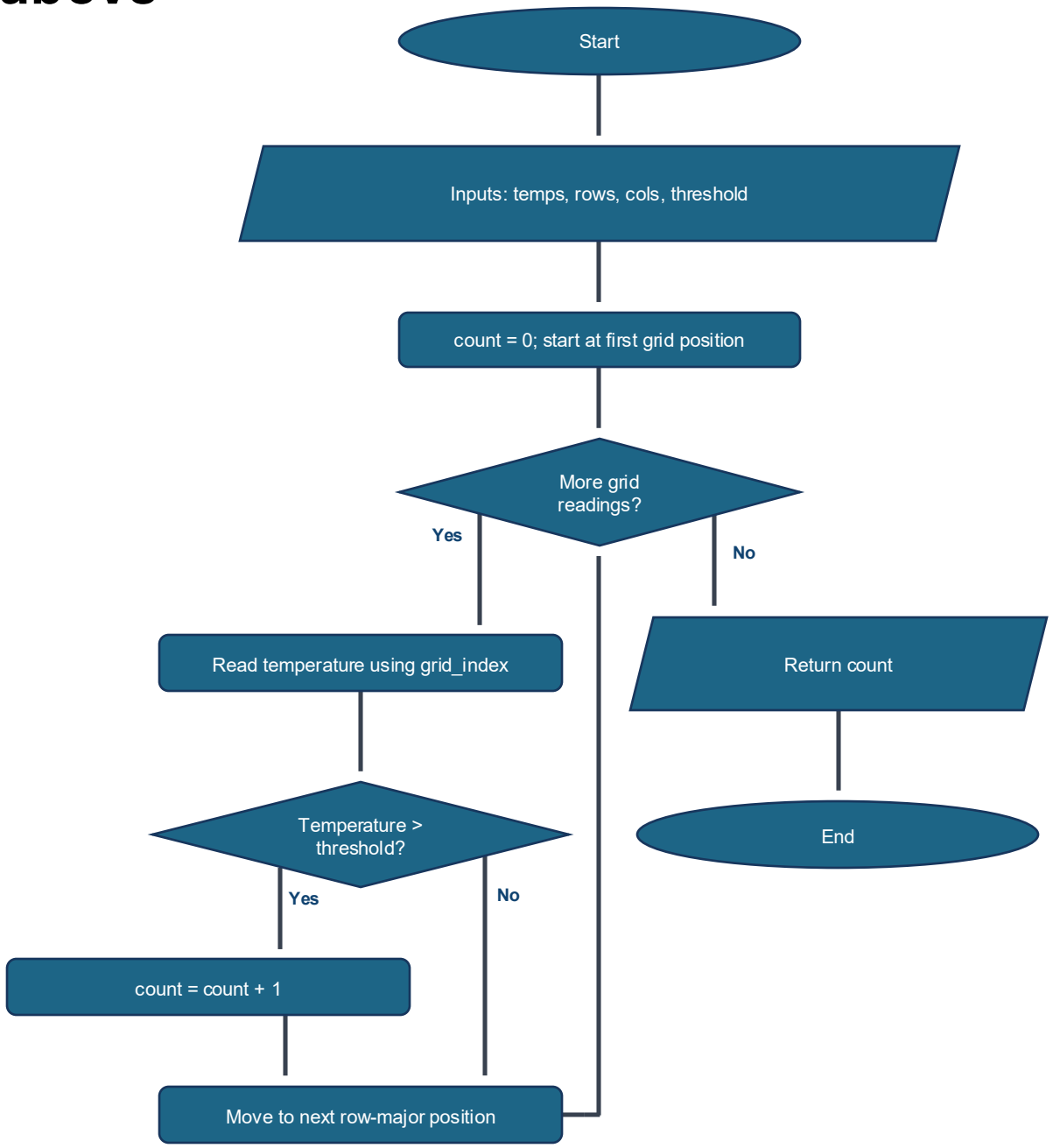
find_hottest



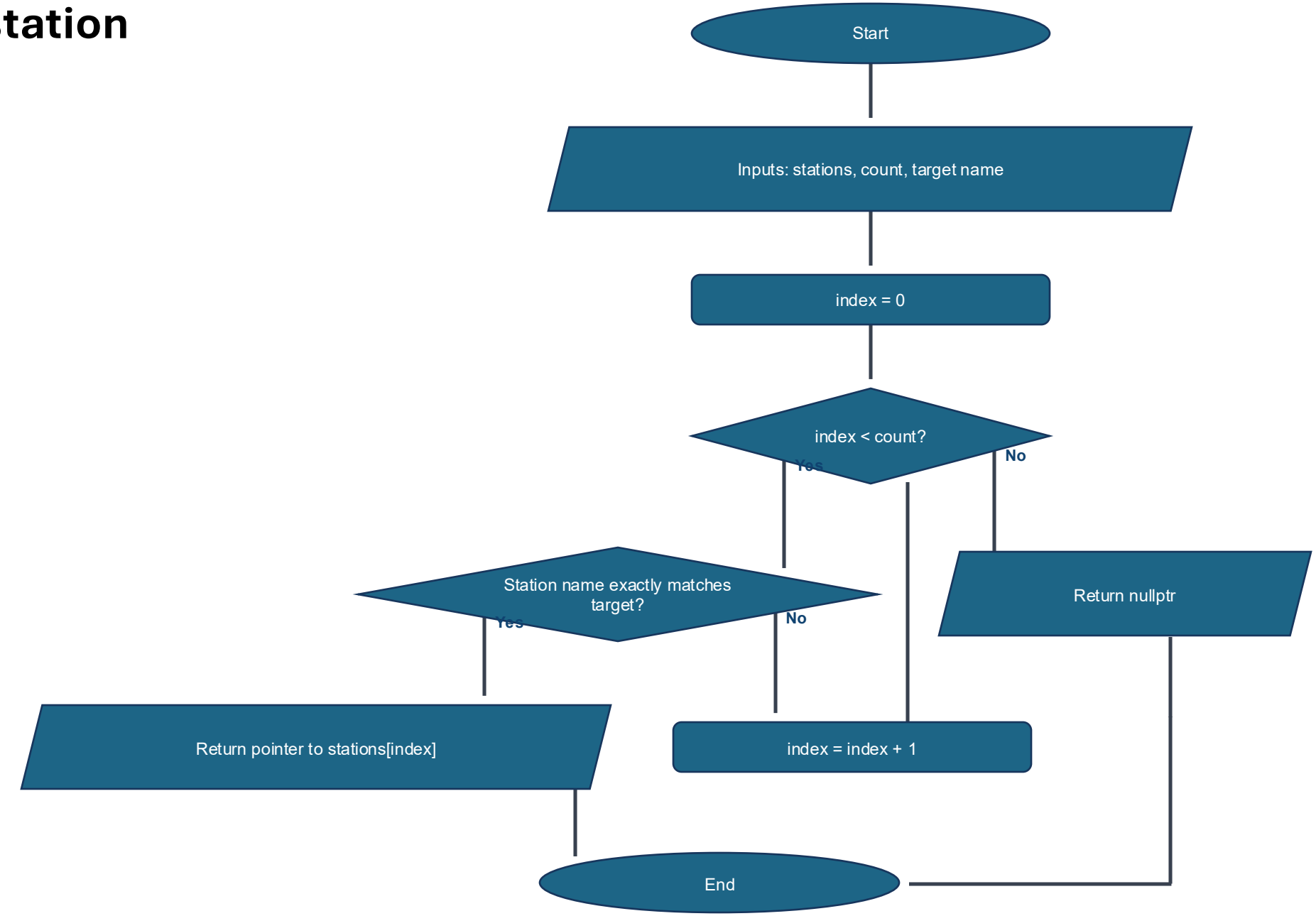
grid_index



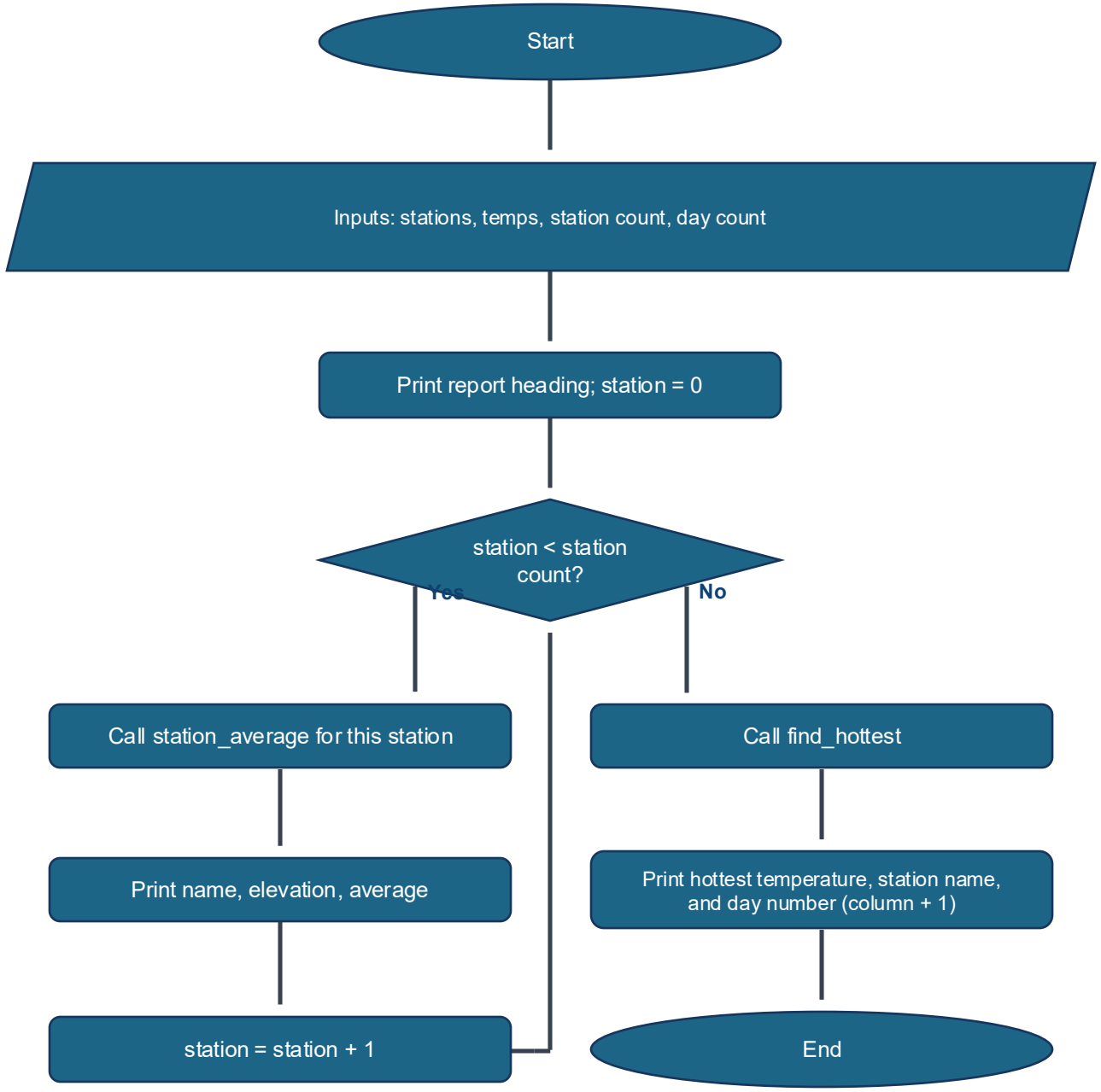
count_above


Start
Inputs: temps, rows, cols, threshold
count = 0; start at first grid position
More grid readings?
Yes
No
Read temperature using grid_index
Return count
Temperature > threshold?
End
Yes
No
count = count + 1
Move to next row-major position

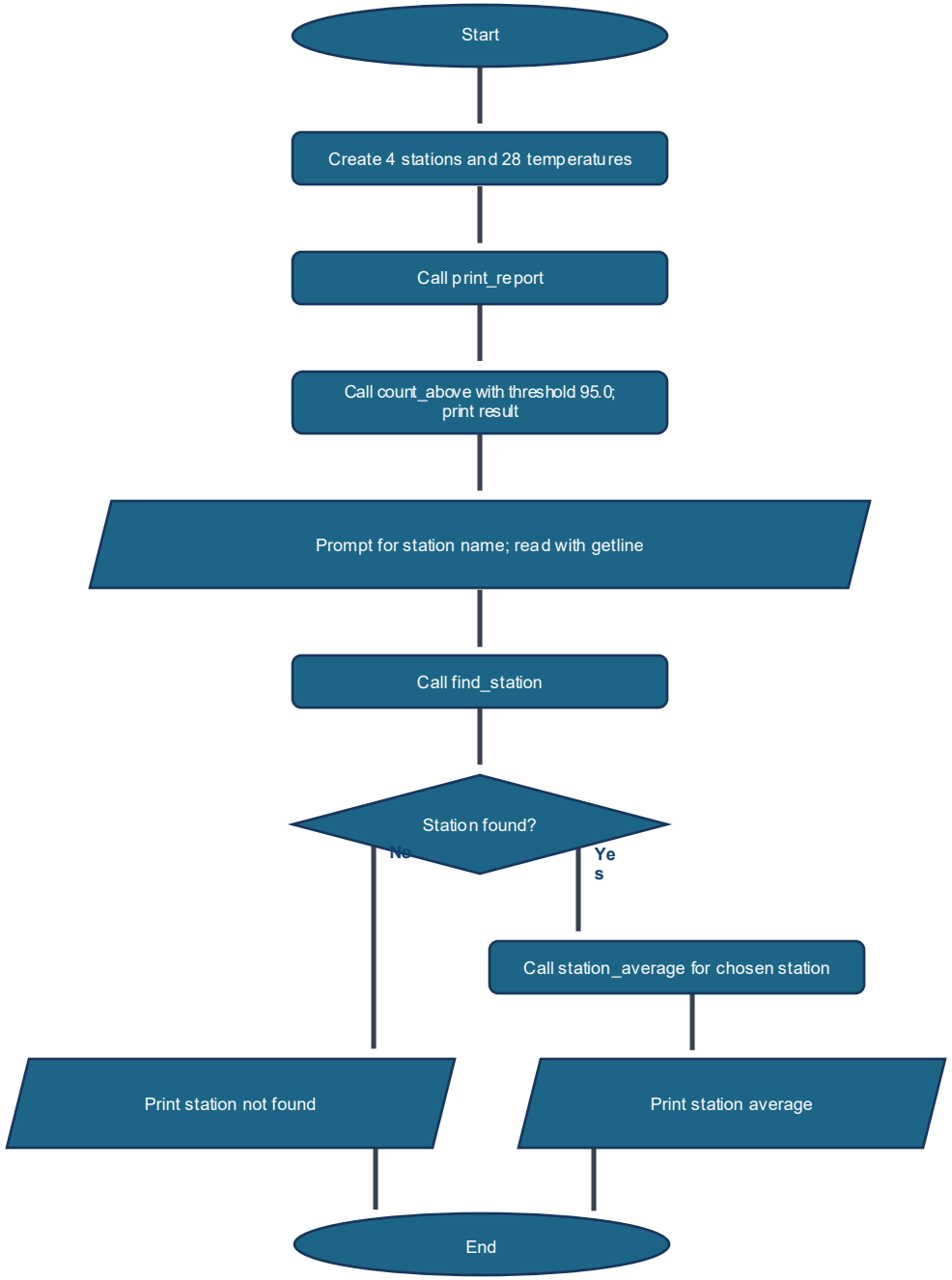
find_station



print_report



main


Start
Create 4 stations and 28 temperatures
Call print_report
Call count_above with threshold 95.0; print result
Prompt for station name; read with getline
Call find_station
Station found?
Ye s
Call station_average for chosen station
Print station not found
Print station average
End